

Experiment 3

Dimensional Modeling and Star Schema Design using Microsoft Power BI

Name: Soumik Roy

UID: 24BAI71017

SUB: Business Intelligence

Aim

To design and implement a dimensional data model using Microsoft Power BI by transforming transactional data from the Global Superstore dataset into a Star Schema consisting of Fact and Dimension tables for efficient reporting and analytical processing.

Tools / Software Required

1. Microsoft Power BI Desktop
2. Global Superstore Dataset.xlsx
3. Power Query Editor (Built into Power BI Desktop)
4. DAX (Data Analysis Expressions)

Theory

Dimensional Modeling is a data warehouse design technique used in Business Intelligence systems to support analytical processing. It organizes data into Fact Tables and Dimension Tables.

Fact Tables contain measurable business events such as Sales, Profit, Quantity, and Discount.

Dimension Tables contain descriptive attributes such as Customer, Product, Region, and Date.

A Star Schema consists of a central Fact Table connected to multiple Dimension Tables. It improves query performance, simplifies reporting, and enhances scalability in BI environments.

Conduct

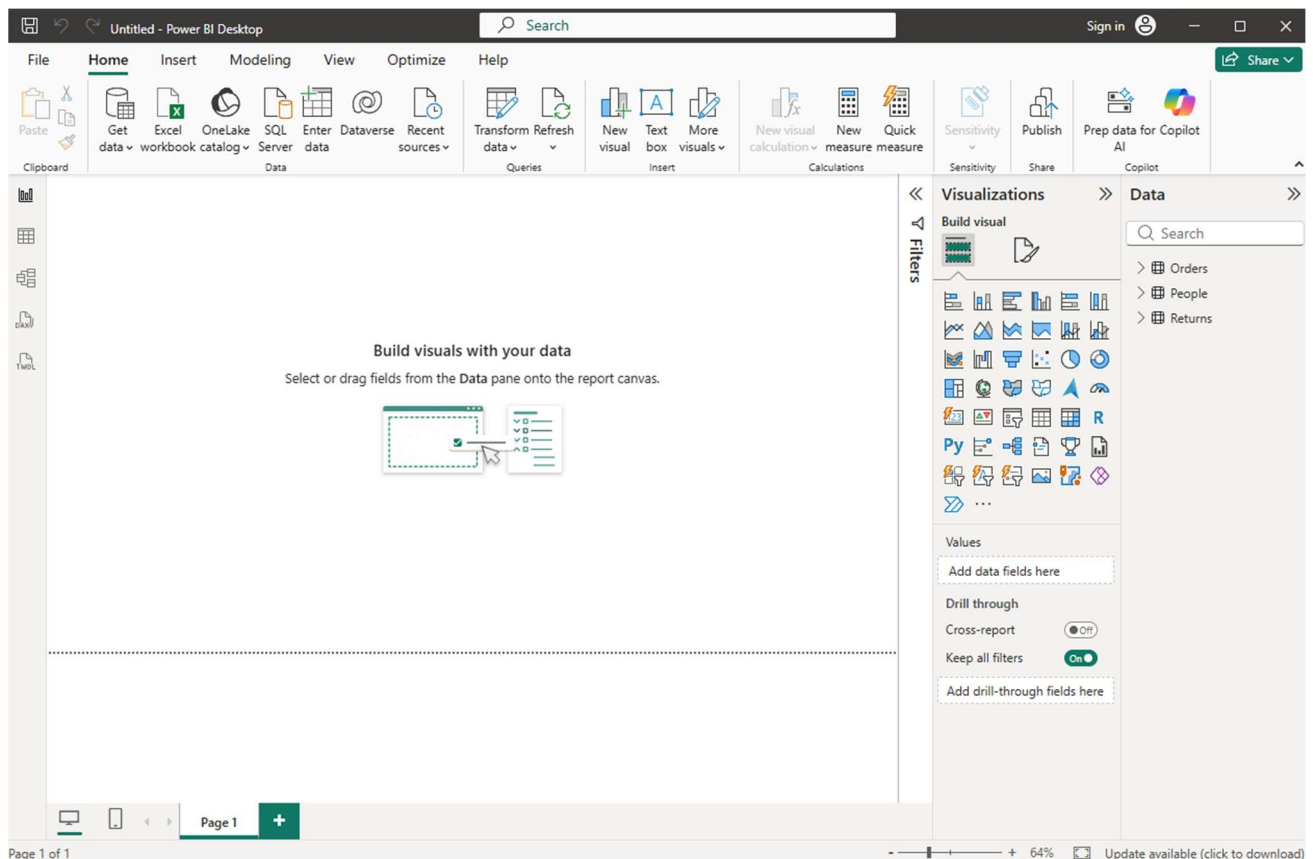
Conduct - Part A: Load Dataset

Step 1: Open Power BI Desktop.

Step 2: Import Global Superstore Dataset.xlsx.

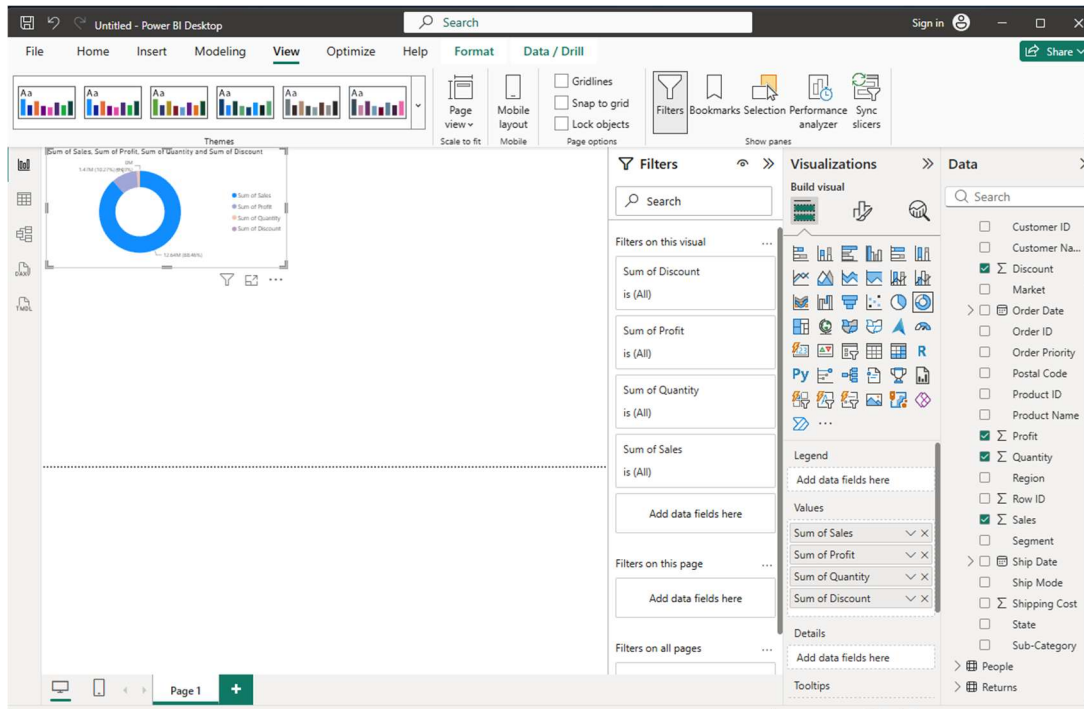
Step 3: Select Orders, Returns, and People worksheets.

Step 4: Click Load and verify imported tables.



Conduct - Part B: Identify Fact and Dimension Tables

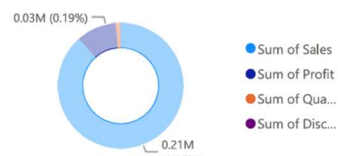
Step 5: Analyze Orders table and identify measures such as Sales, Profit, Quantity, and Discount.



Step 6: Identify descriptive attributes including Customer, Product, Region, and Date information.

Step 7: Categorize attributes into Fact and Dimension tables.

Sum of Sales, Sum of Profit, Sum of Quantity and Sum of Discount



Country and Region

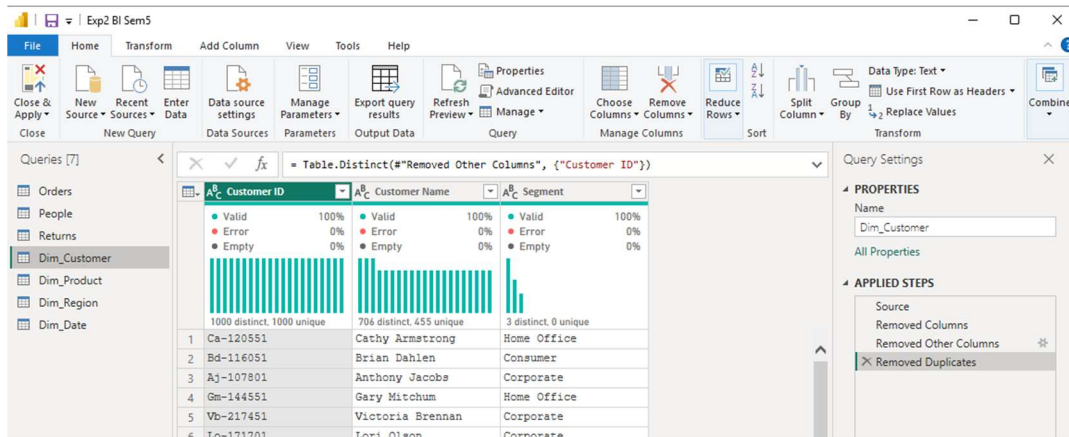


Order ID	Customer ID	Product ID	Region	Year	Quarter	Month
AG-2014-AH2103-41863	AH-2103	TEC-PH-3818	Nor	2014	Qtr 1	January
AG-2014-DH36753-41866	DH-36753	OFF-BI-4807	Nor	2014	Qtr 1	February
AG-2014-DH36753-41866	DH-36753	OFF-PA-5849	Nor	2014	Qtr 1	March
AG-2014-TT110703-41863	TT-110703	OFF-FA-6203	Nor	2014	Qtr 2	April
AU-2014-DS30308-41868	DS-30308	OFF-EN-5025	Wes	2014	Qtr 2	May
BO-2014-DK283513-41879	DK-283513	OFF-BI-4804	East	2014	Qtr 2	June
BO-2014-ES402013-41877	ES-402013	FUR-CH-5800	East	2014	Qtr 3	July
BO-2014-ES402013-41877	ES-402013	OFF-BI-2918	East	2014	Qtr 3	August
BO-2014-SS1059013-41862	SS-1059013	OFF-FA-6190	East	2014	Qtr 3	September
BO-2014-SS1059013-41862	SS-1059013	OFF-SU-4978	East	2014	Qtr 4	October
BU-2014-ME772519-41856	ME-772519	FUR-FU-6257	East	2014	Qtr 4	November
BU-2014-ME772519-41856	ME-772519	OFF-FA-6191	East	2014	Qtr 4	December
CA-2014-AG10525140-41859	AG-105251408	OFF-PA-6572	Sou	2015	Qtr 1	January
CA-2014-AG10525140-41859	AG-105251408	TEC-PH-3160	Sou	2015	Qtr 1	February
CA-2014-AH19523-41859	AH-19523	OFF-EN-3089	Can	2015	Qtr 1	March
CA-2014-AP10915140-41873	AP-109151404	FUR-FU-5679	Wes	2015	Qtr 2	April
CA-2014-AP10915140-41873	AP-109151404	OFF-PA-6485	Wes	2015	Qtr 2	May
CA-2014-AP10915140-41873	AP-109151404	OFF-PA-6533	Wes	2015	Qtr 2	June
CA-2014-AP10915140-41873	AP-109151404	OFF-PA-6546	Wes	2015	Qtr 3	July
CA-2014-AP10915140-41873	AP-109151404	OFF-SU-3000	Wes	2015	Qtr 3	August
CA-2014-AP10915140-41873	AP-109151404	TEC-PH-3814	Wes	2015	Qtr 3	September
CA-2014-AT10735140-41880	AT-107351406	OFF-BI-6410	East	2015	Qtr 4	October
Total				2015	Qtr 4	November

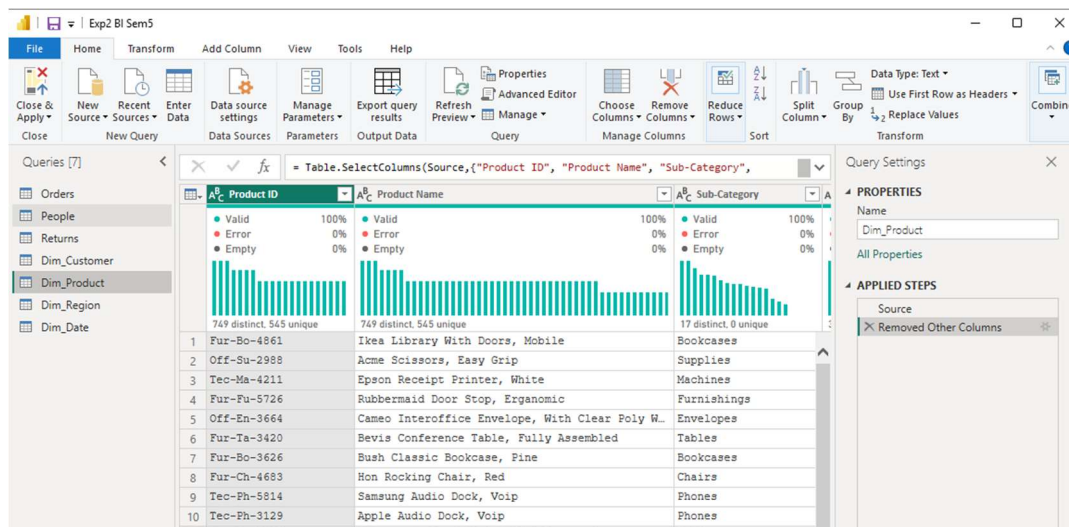
Product ID	Product Name	Customer ID	Customer Name	Segment
FUR-BO-3616	Bush 3-Shelf Cabinet, Mobile	ZD-219251404	Zuschuss Donatelli	Consumer
FUR-BO-3617	Bush 3-Shelf Cabinet, Pine	YS-11880134	Yana Sorensen	Corporate
FUR-BO-3618	Bush 3-Shelf Cabinet, Tradition	YC-1189522	Yoseph Carroll	Corporate
FUR-BO-3625	Bush Classic Bookcase, Mobile	VW-2177544	Victoria Wilson	Corporate
FUR-BO-3635	Bush Floating Shelf Set, Mobile	VP-2173058	Victor Preis	Home Office
FUR-BO-3637	Bush Floating Shelf Set, Tradition	VP-2173045	Victor Preis	Home Office
FUR-BO-3639	Bush Library with Doors, Metal	VG-2180558	Vivek Grady	Corporate
FUR-BO-3642	Bush Library with Doors, Tradition	VG-217901408	Vivek Gonzalez	Consumer
FUR-BO-3647	Bush Stackable Bookrack, Pine	VF-21715102	Vicky Freymann	Home Office
FUR-BO-3888	Dania 3-Shelf Cabinet, Tradition	VD-216701404	Valerie Dominguez	Consumer
FUR-BO-3891	Dania Classic Bookcase, Pine	TZ-214457	Tom Zandusky	Corporate
FUR-BO-3894	Dania Corner Shelving, Mobile	TZ-21445113	Tom Zandusky	Corporate
FUR-BO-3896	Dania Corner Shelving, Tradition	TW-2102527	Tamara Willingham	Home Office
FUR-BO-3897	Dania Floating Shelf Set, Metal	TW-1102568	Tamara Willingham	Home Office
FUR-BO-3901	Dania Library with Doors, Metal	TT-110703	Ted Trevino	Consumer
FUR-BO-3902	Dania Library with Doors, Mobile	TS-2165591	Trudy Schmidt	Consumer
FUR-BO-3907	Dania Stackable Bookrack, Tradition	TS-2161082	Troy Staebel	Consumer
FUR-BO-4667	Hon Metal Bookcases, Gray	TS-2161058	Troy Staebel	Consumer
FUR-BO-4844	Ikea 3-Shelf Cabinet, Metal	TS-216101406	Troy Staebel	Consumer
FUR-BO-4846	Ikea 3-Shelf Cabinet, Pine	TS-21205139	Thomas Seio	Corporate
FUR-BO-4852	Ikea Corner Shelving, Metal	TP-2141592	Tom Prescott	Consumer
FUR-BO-4853	Ikea Corner Shelving, Mobile	TM-2101064	Tamara Manning	Consumer
FUR-BO-4856	Ikea Floating Shelf Set, Metal	TM-210101404	Tamara Manning	Consumer
FUR-BO-4858	Ikea Floating Shelf Set, Pine	TM-21010125	Tamara Manning	Consumer
FUR-BO-4861	Ikea Library with Doors, Mobile	TH-212357	Tiffany House	Corporate
FUR-BO-5520	O'Sullivan 2-Shelf Heavy-Duty B	TH-21100118	Thea Hendricks	Consumer
FUR-BO-5797	Safco Stackable Bookrack, Mobile	TH-1111533	Thea Hudgings	Corporate
FUR-BO-5948	Sauder Classic Bookcase, Metal	TG-213108	Toby Gnade	Consumer
FUR-BO-5950	Sauder Classic Bookcase, Pine	TG-2131045	Toby Gnade	Consumer
FUR-BO-5952	Sauder Corner Shelving, Metal	TG-2131018	Toby Gnade	Consumer
FUR-BO-5969	Sauder Library with Doors, Tradition	TC-2129527	Toby Carlisle	Consumer
FUR-BO-5972	Sauder Stackable Bookrack, Pine	TB-2162582	Trudy Brown	Consumer
FUR-CH-4415	Global Leather and Oak Executive	TB-216251406	Trudy Brown	Consumer
FUR-CH-4428	Global Value Steno Chair, Gray	TB-2125045	Tim Brockman	Consumer

Conduct - Part C: Create Dimension Tables

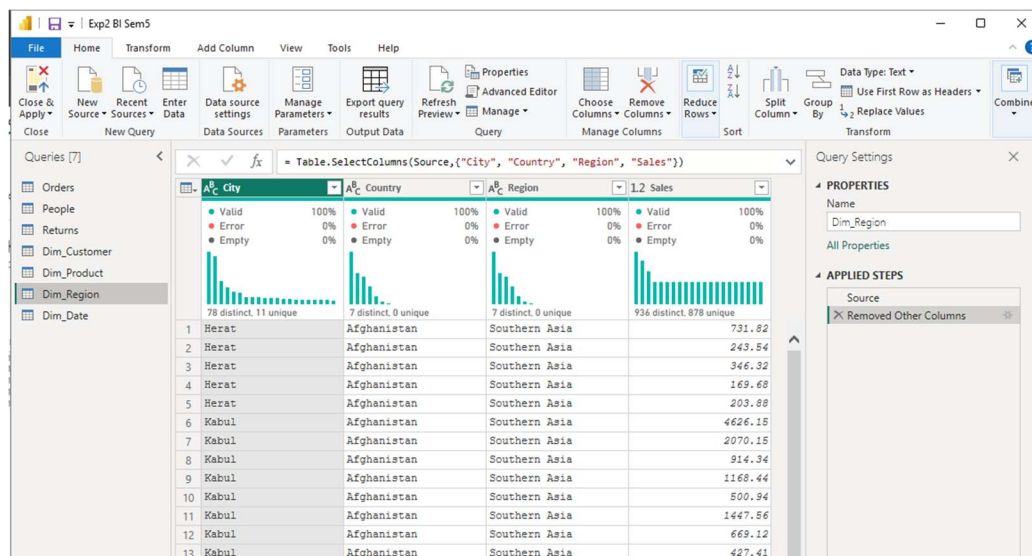
Step 8: Create Dim_Customer using Customer ID, Customer Name, and Segment.

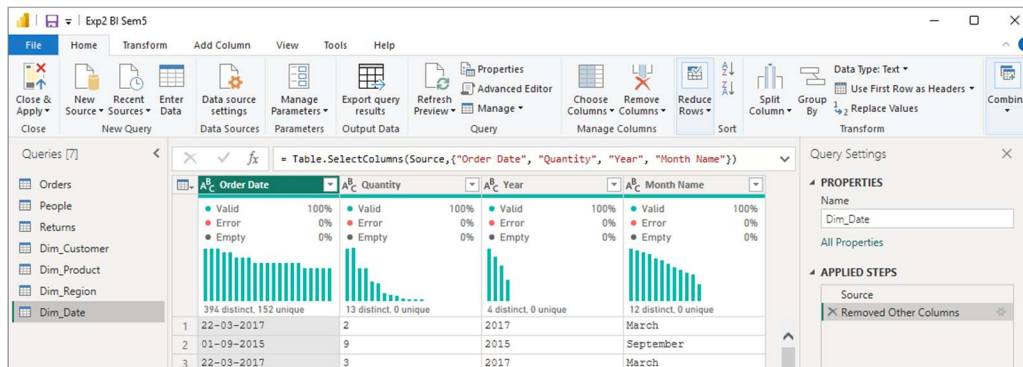
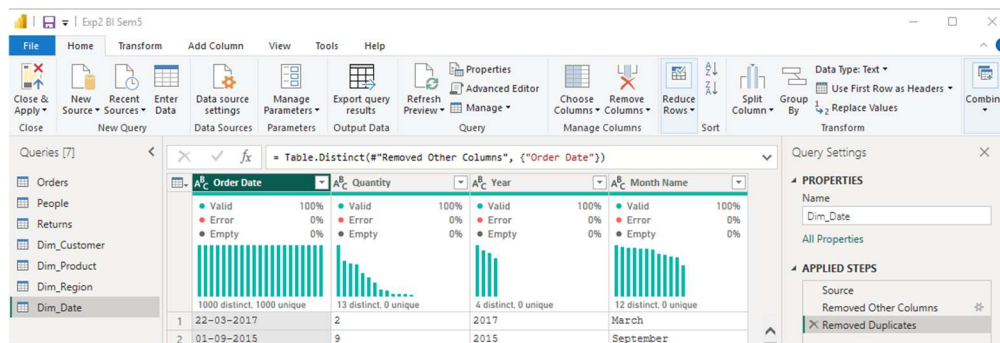


Step 9: Create Dim_Product using Product ID, Product Name, Category, and Sub-Category.



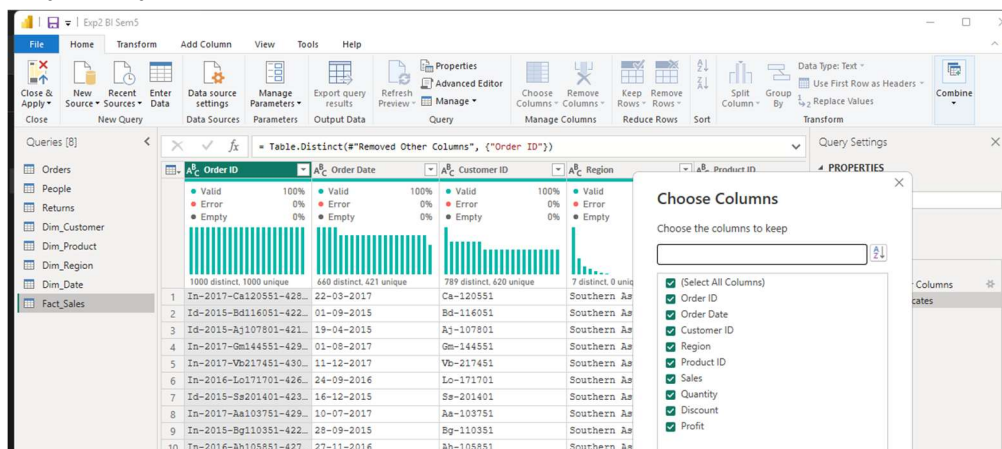
Step 10: Create Dim_Region using Region, Country, State, and City.



Step 11: Create Dim_Date using Order Date, Year, Quarter, and Month.**Step 12: Remove duplicate records from all dimensions.****Conduct - Part D: Create Fact Table**

Step 13: Create Fact_Sales containing Order ID, Customer ID, Product ID, Region, Order Date, Sales, Quantity, Profit, and Discount.

Step 14: Verify all key columns required for relationships.



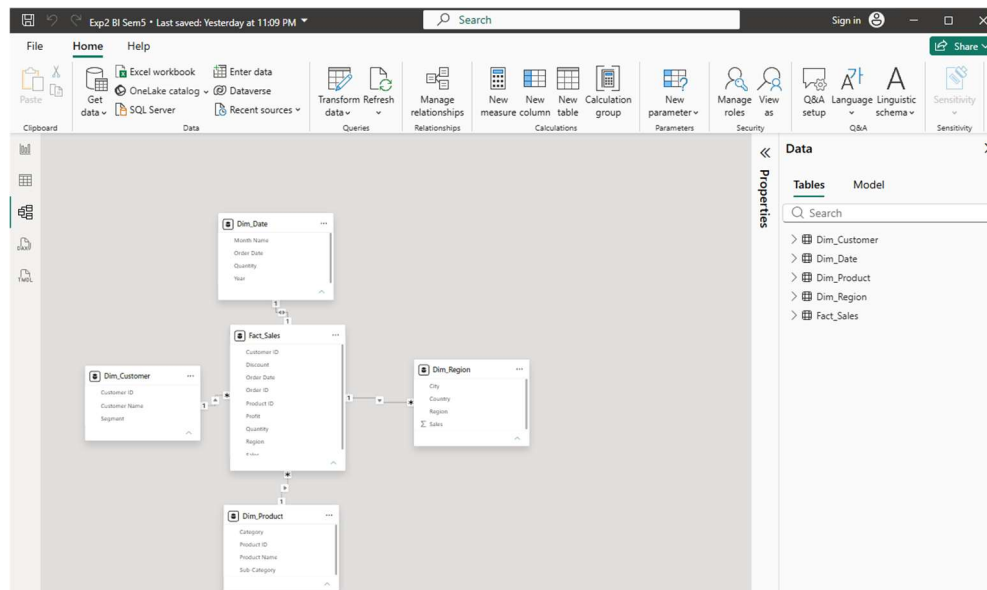
Conduct - Part E: Build Star Schema

Step 15: Open Model View.

Step 16: Create relationships between Fact_Sales and all Dimension Tables.

Step 17: Configure relationships as Many-to-One.

Step 18: Arrange tables visually into a Star Schema layout.

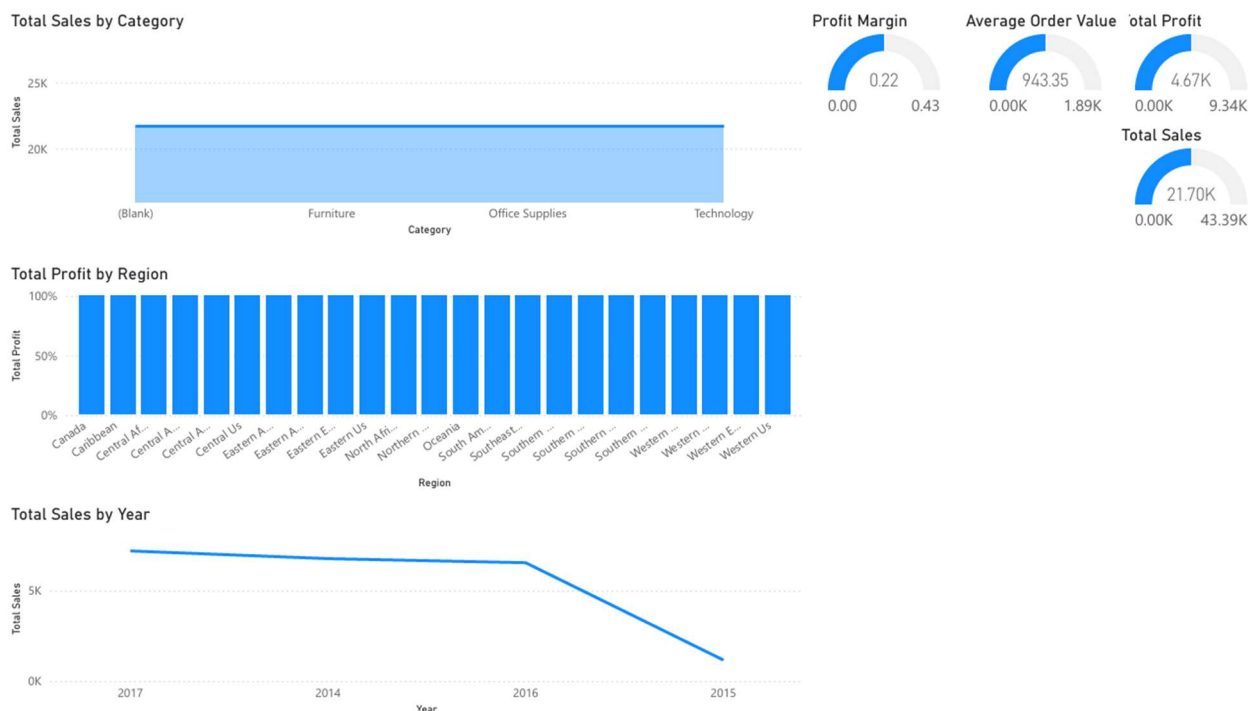


Conduct - Part F: Validate Model

Step 19: Create visualizations such as Sales by Category, Profit by Region, and Sales Trend by Year.

Step 20: Create DAX measures including Total Sales, Total Profit, Profit Margin, and Average Order Value.

Step 21: Verify that dimensions correctly filter fact data.



Result

A dimensional model was successfully designed and implemented using Microsoft Power BI. A Star Schema was created consisting of one Fact Table and multiple Dimension Tables. Relationships and DAX measures were successfully implemented and validated using reports.

Learning Outcomes

- Explain dimensional modeling concepts and Star Schema architecture. (CO2-BT4)
- Differentiate between Fact and Dimension tables. (CO2-BT4)
- Design dimensional models using real-world business datasets. (CO5-BT6)
- Create and validate Star Schemas in Power BI. (CO5-BT6)
- Develop DAX measures for analytical reporting. (CO5-BT6)

Text / Reference Books

1. Sharda, Delen & Turban, Business Intelligence and Analytics: Systems for Decision Support.
2. Cindi Howson, Successful Business Intelligence.
3. Microsoft Power BI Official Documentation.
4. Microsoft Power Query Documentation.